QXU5030 Composite Materials

Exam Revision — Checklist Q&A

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Q1 Explain the role of fibre, matrix and interface in a composite. [纤维 / 基体 / 界面的作用]

- Fibre（纤维）: Reinforcing phase. Carries the mechanical load; provides high strength and stiffness. Contribution governed by: (1) basic mechanical properties, (2) surface interaction with matrix (interface), (3) fibre volume fraction Vf, (4) fibre orientation.
- Matrix（基体）: Binds fibres together; transfers and distributes load between fibres; protects fibres from damage; determines processing route and chemical/environmental resistance.
- Interface（界面）: Region between fibre and matrix. Transfers shear stress from matrix to fibre. Strong interface → higher stiffness but more brittle (less pull-out energy). Weak interface → lower strength but tougher. Interface strength needs optimisation, not maximisation.

Q2 Compare glass, carbon and aramid fibres in terms of cost, stiffness, impact resistance and applications. [三种纤维对比]

Property 性能	Glass 玻璃纤维 £1–2/kg	Carbon 碳纤维 £15–40/kg	Aramid 芳纶 £15–25/kg
Cost（成本）	Lowest	High	Medium-high
Stiffness E（模量）	~70 GPa (like Al)	Up to 400+ GPa; 25× steel by weight	LM 60 GPa / HM 125 GPa
Impact resistance（抗冲击）	Moderate	Poor — brittle	Excellent — ballistic use
Compressive strength（抗压）	Good	Good	Poor (~20% of tensile; kink bands)
Applications（应用）	Wind turbines, boats, automotive panels	Aerospace wings, F1, sporting goods	Helmets, body armour, cables, ropes

Q3 Explain why vinyl ester, polyester and epoxy matrices are selected for different service environments. [三种树脂基体的选用]

- Polyester（不饱和聚酯，£1–2/kg）: Reactive sites along chain; cross-links with styrene. Cures at room temperature without pressure. Lowest cost. Many ester groups → susceptible to water hydrolysis（水解）→ poor chemical resistance. Use: boats, wind turbine blades, general FRP.
- Vinyl Ester（乙烯基酯，£2–4/kg）: Reactive sites only at chain ends → more ductile（延展性好）. Fewer ester groups → better water and chemical resistance. Used for chemical storage tanks, pipelines, marine environments where moisture resistance is critical.
- Epoxy（环氧树脂，£3–15/kg）: No ester groups → excellent water resistance. Aromatic rings → high stiffness and thermal resistance (up to 140°C wet / 220°C dry). Cured by amine hardener (addition reaction — correct mix ratio critical). Best overall mechanical properties. Use: aerospace structures, F1 cars, high-end sporting goods.

Q4 Explain the difference between thermosets and thermoplastics, including recyclability and processing limitations. [热固性 vs 热塑性]

Property	Thermoset 热固性（e.g. epoxy, polyester）	Thermoplastic 热塑性（e.g. PEEK, PP, PA）
Molecular structure（分子结构）	3D cross-linked network — irreversible chemical reaction（不可逆固化）	Linear 1D chains — held by intermolecular forces; can be remelted
Cure / processing（加工方式）	Liquid resin + hardener → cross-link. Long cure cycle (hours). Autoclave or RTM.	Melt and reshape. Cycle <1 min possible. BUT high viscosity（高粘度）(10–100 Pa·s vs 0.2–2 Pa·s for TS) makes fibre impregnation difficult.
Recyclability（可回收性）	Very difficult. Matrix lost in pyrolysis（热解）. Fibres can be recovered. EU 95% recyclability target is a major challenge.	Easier in principle — remelt and re-form. Short-fibre TP composites used in injection moulding. Continuous fibre TP also recyclable.
Advantage（优势）	High Vf achievable (60%); superior mechanical & thermal properties; well-established aerospace processes	Tougher matrix; weldable（可焊接）; long shelf life; no solvent; potentially eco-friendly

Q5 Justify why 0°, 90° and ±45° plies are used in laminates. [铺层角度的设计原因]

- 0° plies（纵向铺层）: Carry primary tensile/compressive loads along the fibre direction. Provide maximum axial stiffness E_1 and tensile strength. Essential wherever a known dominant load direction exists.
- 90° plies（横向铺层）: Resist transverse loads; prevent splitting parallel to 0° fibres; provide lateral stiffness E_2. Without them the laminate is very weak transversely.
- ±45° plies（剪切铺层）: Provide in-plane shear stiffness G_{12} and strength. Critical for torsional loading, combined loading and structural stability (buckling prevention).

Quasi-isotropic lay-up [0/90/±45] gives approximately equal modulus in all in-plane directions: $E_{QI} \approx 3/8\ E_0$（约为单向模量的 3/8）. Example: HS-carbon UD E_0 = 110 GPa → quasi-isotropic ≈ 45 GPa. Designer can tailor property direction by adjusting ply ratios.

Q6 Explain why balanced and symmetric laminates reduce twisting and improve reliability. [均衡对称铺层]

- Symmetric laminate（关于中面对称）: Ply stacking is mirror-image about the mid-plane. Eliminates bending-extension coupling (B-matrix = 0). Prevents warping or bowing under in-plane loads or during cooling after cure. Without symmetry, residual thermal stresses cause the panel to spring out of shape after demoulding.
- Balanced laminate（均衡，+θ 与 -θ 成对）: For every ply at +θ there is an equal ply at -θ. Eliminates in-plane shear-extension coupling. Without balance, a tensile load causes the panel to twist.

Combined balanced + symmetric → predictable in-plane behaviour; reduces manufacturing distortion; improves fatigue reliability by avoiding unexpected coupling deformations.

Q7 Select a suitable process for low-volume, high-volume, hollow, constant-section and panel-like components. [工艺选择]

Component type 构件类型	Process 工艺	Key reason 原因
Low-volume 小批量 (1–500)	Hand lay-up / Autoclave prepreg	Low tooling cost; flexible; achieves high Vf (~60%) and aerospace quality
High-volume 大批量 (>5000)	Compression moulding (SMC) / RTM	Cycle time 1–5 min; automated; consistent quality; complex shapes
Hollow 中空 (pipes, tanks)	Filament winding 缠绕成型	Mandrel provides hollow form; fibres wound in optimal pattern; excellent hoop strength
Constant cross-section 等截面 (beams)	Pultrusion 拉挤成型	Continuous automated pull-through die; fast; high Vf; structural profiles
Large panel, one moulded surface	Vacuum infusion (VARTM)	Large area; low tooling cost (one-sided mould + vacuum bag); good fibre wet-out
Complex aerospace part, both surfaces	RTM (Resin Transfer Moulding)	Closed mould; both surfaces moulded; good dimensional control; low void content

Q8 Explain how voids, dry spots, poor cure and fibre misalignment reduce performance. [加工缺陷的影响]

- Voids（孔隙/气泡）: Gas trapped during lay-up or cure. Reduce interlaminar shear strength (ILSS 层间剪切强度) — even 1% void content can reduce ILSS by ~7%. Act as crack initiation sites（裂纹萌生点）; lower fatigue life.
- Dry spots（干斑）: Areas where fibres are not wetted by resin — no matrix present. Zero load transfer in that region; local stress concentration; significantly reduced strength and stiffness at that location.
- Poor cure（固化不足）: Incomplete cross-linking due to wrong hardener ratio, insufficient temperature or time. Results in soft matrix with reduced stiffness and strength; lower glass transition temperature Tg（玻璃化转变温度）; poor moisture and chemical resistance.
- Fibre misalignment（纤维偏角）: Fibres not oriented as designed. Anisotropy means even a 10° off-axis deviation drastically reduces tensile modulus. Also introduces unexpected coupling deformations not present in the design.

Q9 Calculate or explain E_1, E_2 and G_{12} using rule-of-mixtures ideas. [混合律 E_1, E_2, G_{12} 推导]

Assumptions（假设）: fibres are uniform, parallel, continuous; perfect bonding at interface; for longitudinal case — equal strain in fibre and matrix.

$E_1 = Ef \cdot Vf + Em \cdot (1-Vf)$	Voigt model — longitudinal / axial modulus（纵向模量）. Equal strain assumption（等应变）. Upper bound（上界）. Accurate for E_1.
$1/E_2 = Vf/Ef + Vm/Em$	Reuss model — transverse modulus（横向模量）. Equal stress assumption（等应力）. Lower bound（下界）. Less accurate: matrix near fibres is constrained, so true E_2 > predicted. If Ef >> Em: $E_2 \approx Em / Vm$
$1/G_{12} = Vf/Gf + Vm/Gm$	Shear modulus (in-plane)（面内剪切模量）. Reuss-type inverse ROM. Significantly under-estimates G_{12}. Halpin-Tsai gives better result.

Where: Vf = fibre volume fraction（纤维体积分数）; Vm = 1 - Vf; Ef, Em = fibre and matrix Young's moduli; Gf, Gm = shear moduli.

Q10 Explain short-fibre stress transfer, critical fibre length, fibre orientation and interface load transfer. [短纤维应力传递与临界长度]

- Stress transfer mechanism（应力传递）: In short-fibre composites, load transfers from matrix to fibre via interfacial shear stress τ_i (shear lag model 剪切滞后). Tensile stress in the fibre builds from zero at the fibre ends to a maximum at the centre. Matrix carries higher stress near fibre ends.

$L_c = \sigma_f \cdot d_f / (2\tau_i)$	Critical fibre length（临界纤维长度）. σ_f = fibre tensile strength; d_f = fibre diameter（纤维直径）; τ_i = interfacial shear strength. Derived by balancing tensile force on fibre cross-section vs. shear force along interface.

- L > Lc (supercritical, 超临界): Fibre reaches its full tensile strength — efficient reinforcement. Fibre breaks before pull-out.
- L < Lc (subcritical, 次临界): Fibre cannot develop full stress — inefficient. Matrix or interface fails first; fibre pulls out intact.
- Fibre orientation effect（取向效率）: Aligned (0°): orientation efficiency $\eta = 1$. Random in-plane: $\eta \approx 3/8$. Random 3D: $\eta \approx 1/5$. Misalignment greatly reduces effective composite modulus.

Q11 Explain wetting, interfacial bonding, debonding, matrix cracking and fibre pull-out. [界面现象与失效机制]

- Wetting（润湿）: Resin must spread over the fibre surface to form intimate contact (low contact angle). Prerequisite for any adhesion. Surface treatment (sizing 浸润剂, oxidation of CF) improves wettability.
- Interfacial bonding（界面结合）: Achieved by: physical (van der Waals), chemical (covalent/ionic via coupling agents), or mechanical (surface roughness interlocking) adhesion. Carbon fibre receives surface oxidation + sizing to improve chemical bonding with epoxy.
- Debonding（界面脱黏）: When interfacial shear stress exceeds IFSS（界面剪切强度）, the fibre separates from the matrix locally. First damage mode in transverse loading. Initiates at stress concentrations.
- Matrix cracking（基体开裂）: Transverse matrix cracks initiate in 90° plies at ~0.3% strain. Multiple cracking occurs with increasing load until saturation. Does not cause immediate catastrophic failure — fibres in 0° plies still carry load.
- Fibre pull-out（纤维拔出）: When a fibre breaks and its broken end slides out of the matrix. Energy absorbed by friction along the debonded interface — key toughening mechanism（韧化机制）. High pull-out length ↔ weak interface ↔ more toughness but lower stiffness/strength. Visible as white fracture surface.

Q12 Use evidence from inspection and service conditions to justify retain, repair or replace decisions. [保留 / 修复 / 替换 决策框架]

Decision depends on: damage type, extent, location (critical vs non-critical load path 承载路径), residual strength vs design allowable, and cost-effectiveness of repair.

Decision	Evidence / Conditions 依据	Justification 理由
RETAIN 保留	Surface scratch or cosmetic damage only; no structural fibres broken; residual strength > design allowable; no delamination detected by C-scan or tap test	Damage does not compromise load path; structural integrity maintained; monitoring sufficient; no action cost-effective
REPAIR 修复	Localised delamination（分层）or small impact damage; fibres partially intact in non-primary load area; repairable geometry accessible; damage within repair manual allowable	Patch repair (wet lay-up or prepreg patch) restores local stiffness; seals against moisture ingress; extends component life at lower cost than replacement. CFRP composites have poor repairability compared to metals — this is a key limitation.
REPLACE 替换	Through-thickness fibre fracture in primary load-bearing region; BVID (Barely Visible Impact Damage 勉强可见冲击损伤) in critical zone; large delamination beyond repair allowable; matrix chemical degradation; fatigue crack growth beyond safe limit	Structural integrity cannot be restored by repair; failure risk exceeds safety margin; continued service would be unsafe or unairworthy

- Key inspection methods（检测手段）: Visual inspection（目视检查）; tap test（敲击测试）; ultrasonic C-scan（超声C扫描）; X-ray; thermography（红外热像仪）. Always compare residual strength against design load requirement.

Quick Key-Term Reference 重点词汇速查

English Term	中文	One-line reminder
Fibre volume fraction Vf	纤维体积分数	Vf = Volume_fibre / Volume_composite
Rule of Mixtures (Voigt)	混合律（上界）	$E_1 = Ef \cdot Vf + Em \cdot Vm$ — equal strain, longitudinal
Inverse ROM (Reuss)	逆混合律（下界）	$1/E_2 = Vf/Ef + Vm/Em$ — equal stress, transverse
Critical fibre length Lc	临界纤维长度	$Lc = \sigma f \cdot df / 2\tau_i$
IFSS	界面剪切强度	Measured by pull-out, microbond, ILSS (short-beam shear)
Wetting / Sizing	润湿 / 浸润剂	Resin spreading over fibre; sizing improves adhesion
Debonding	界面脱黏	Interfacial failure when $\tau >$ IFSS
Fibre pull-out	纤维拔出	Key toughening mechanism — frictional energy absorption
Matrix cracking	基体开裂	Initiation at ~0.3% strain in 90° plies of cross-ply laminates
Delamination	分层	Inter-ply crack driven by interlaminar shear / impact
Void content	孔隙率	~1% voids → ~7% reduction in ILSS
Quasi-isotropic lay-up	准各向同性铺层	[0/90/±45]; $E \approx 3/8\ E_0$ in all in-plane directions
Symmetric laminate	对称铺层	No bending-extension coupling; prevents warping
Balanced laminate	均衡铺层	+θ and -θ in equal pairs; no shear-extension coupling
Hand lay-up	手工铺层	Low-volume; room-temp cure; skill-dependent
Autoclave	热压罐固化	High Vf (60%); aerospace quality; expensive & slow
RTM	树脂传递成型	Closed mould; good both surfaces; small-medium parts
Filament winding	纤维缠绕	Hollow/cylindrical; mandrel; pipes and tanks
Pultrusion	拉挤成型	Constant cross-section; continuous; beams and profiles
SMC + Compression	片状模塑料+压缩成型	High-volume thermoset; 1–5 min cycle; automotive
Thermoset	热固性树脂	3D cross-linked; irreversible cure; hard to recycle
Thermoplastic	热塑性树脂	Remeltable; recyclable; high viscosity processing challenge
Pyrolysis	热解（回收工艺）	Thermal decomposition of TS matrix to recover fibres
BVID	勉强可见冲击损伤	Critical barely-visible damage → often triggers Replace decision
ILSS	层间剪切强度	Measured by short-beam shear test; sensitive to voids
Tg	玻璃化转变温度	Above Tg, matrix softens; poor cure lowers Tg
Synergism / Composite effect	协同效应/复合效应	Observed property >> Rule-of-Mixtures prediction (e.g. toughness)
Anisotropy	各向异性	Properties direction-dependent; key feature of composites